외계행성 대기
Exoplanetary Atmospheres

Lecture 8: Vertical Thermal Structure of a Planetary Atmosphere II
2026 November 2 (Monday)

Based on: Chapter 9 of Seager, *Exoplanet Atmospheres* (2010), and Chapter 4 of Heng, *Exoplanetary Atmospheres: Theoretical Concepts and Foundations* (2017)

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Outline – Lecture 8

Surface Temperature in Simple Greenhouse Models (Continued)

Convection versus Radiation

The Radiative-Equilibrium Temperature Profile

The Adiabatic Temperature Profile

Summary

A leaky greenhouse: the setup

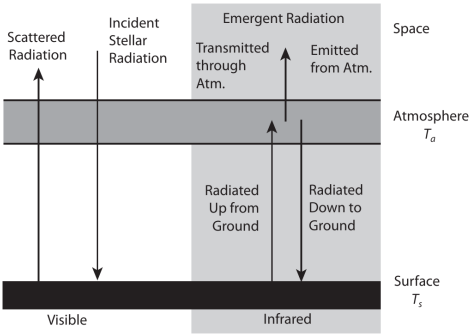


Figure 1: The leaky greenhouse: the layer absorbs only a fraction α of the infrared. Seager (2010), Fig. 9.5.

The simple greenhouse atmosphere traps *all* infrared radiation from the ground. A slightly more realistic "leaky" layer is partly transparent (optically thin) in the infrared:

- the atmosphere layer absorbs a fraction $\alpha \in (0,1)$ of the upwelling surface emission and transmits $1-\alpha$ to space;
- It emits the flux $\mathscr{F}_{\text{atm}}\uparrow = \sigma T_a^4$ upward and $\mathscr{F}_{\text{atm}}\downarrow = \sigma T_a^4$ downward;
- starlight is again assumed to pass the layer and heat the ground. Again, the starlight is assumed not to be scattered back to space and all of the radiation reaches the planet surface.

A leaky greenhouse: the solution

Two contributions now leave the planet (Fig. 9.5):

- emission from the atmosphere layer;
- surface emission that **leaks** through the layer.

Two energy-balance equations (top and surface) determine the two unknowns T_s and T_a:

$$\begin{aligned}
\text{top of the atmosphere:} \quad & \mathscr{F}_*\downarrow & &= \mathscr{F}_{\text{atm}}\uparrow + (1-\alpha)\,\sigma T_s^4\uparrow, \\
\text{surface:} \quad & \mathscr{F}_*\downarrow & + \mathscr{F}_{\text{atm}}\downarrow &= \sigma T_s^4\uparrow, \\
\text{in atmosphere:} \quad & \mathscr{F}_{\text{atm}}\uparrow & + \mathscr{F}_{\text{atm}}\downarrow &= \alpha\sigma T_s^4\uparrow .
\end{aligned}$$

Here, $\mathscr{F}_*\downarrow = \sigma T_e^4$ is the stellar flux absorbed by the ground and $\mathscr{F}_{\text{atm}}\uparrow = \mathscr{F}_{\text{atm}}\downarrow = \sigma T_a^4$ is the flux emitted by the atmosphere layer.
Eliminating $\mathscr{F}_{\text{atm}}\downarrow = \mathscr{F}_{\text{atm}}\uparrow$: $2T_e^4 = (2-\alpha)\,T_s^4$. Then $T_a^4 = T_s^4 - T_e^4 = \alpha T_e^4/(2-\alpha)$:

$$\boxed{T_s = \left(\frac{2}{2-\alpha}\right)^{1/4} T_e}, \qquad \boxed{T_a = \left(\frac{\alpha}{2-\alpha}\right)^{1/4} T_e = \left(\frac{\alpha}{2}\right)^{1/4} T_s}.$$

Check: the atmosphere layer absorbs $\alpha\sigma T_s^4$ and emits $2\sigma T_a^4$, so $T_s^4 = (2/\alpha)T_a^4$.

A leaky greenhouse: what it means

$$T_s = \left(\frac{2}{2-\alpha}\right)^{1/4} T_e, \qquad T_a = \left(\frac{\alpha}{2-\alpha}\right)^{1/4} T_e$$

- **A partly transparent layer reduces the greenhouse warming.**
- $\alpha = 1$: the simple greenhouse, $T_s = 2^{1/4} T_e$ (a special case of the leaky model).
- $\alpha \to 0$: transparent layer or no atmosphere, $T_s = T_e$ and $T_a = 0$ (no atmosphere).
- Since $1 \le 2-\alpha \le 2$: always $\boxed{T_s \ge T_e \ge T_a}$. The ground is warmer than the emission temperature, and the atmospheric layer is colder than it.
- Earth: $T_s = 288\,\mathrm{K}$ with $T_e = 255\,\mathrm{K}$ needs $2/(2-\alpha) = (288/255)^4 \approx 1.63$, i.e. $\alpha \approx 0.77$.
- **A single layer can never give more than $T_s = 2^{1/4} T_e$. Venus needs many opaque layers** (next slide). Extended to many layers, with the right α for each layer at every wavelength, and with molecules and clouds, the model approaches a multilayer **radiative-equilibrium** model.

Many ‘non-leaky’ layers: towards radiative equilibrium

Stack n layers above the surface (Seager’s exercise 3). Assumptions:

- All layers are **identical in their optical properties**: transparent in the visible (starlight heats only the ground) and **fully opaque in the infrared** ($\alpha = 1$), each emitting as a black body up and down.
- Only radiation carries energy (no scattering, convection or conduction). Thickness, density and pressure never enter; the layers differ *only* in temperature, which is an output of energy balance, not an input.

Number the layers $k = 1, \ldots, n$ from the top, with the surface as $k = n+1$ and $T_0 = 0$ (no infrared from space). Layer k absorbs the emission of its two neighbors and emits from both faces, $2\sigma T_k^4 = \sigma T_{k-1}^4 + \sigma T_{k+1}^4$, with $\sigma T_1^4 = \sigma T_e^4$ (the top layer is what space sees). T_k^4 is therefore linear in k:

$$T_k^4 = k\,T_e^4 \quad \Longrightarrow \quad \boxed{T_s = (n+1)^{1/4}\,T_e} \quad \text{(for } n \text{ opaque layers with } \tau_k \approx 1 \text{ and } \tau_{\text{tot}} \approx n\text{)}.$$

- Venus: $(T_s/T_e)^4 = (730/230)^4 \approx 100$, i.e. the equivalent of ~ 100 opaque layers — an infrared optical depth of order 100 above the surface. The massive CO_2 atmosphere (92 bar) is that opaque.
- With $n \to$ many layers of appropriate absorptivity, at all wavelengths, with molecules and clouds, these models become a **multilayer radiative-equilibrium model**: the subject of Lecture 8.
- Each “layer” is one unit of infrared optical depth, not a fixed geometric thickness. The model is a discrete gray atmosphere with $\tau \approx n$, and $T_k^4 \propto k$ mirrors the gray $T^4 \propto \tau$.
- For leaky layers ($\alpha < 1$) the recursion changes and $T_s < (n+1)^{1/4} T_e$; already for $n = 1$, $T_s^4 = 2T_e^4/(2-\alpha)$.
- The one-layer ratio $T_a/T_s = 2^{-1/4}$ reappears in the gray model as the skin temperature $T(0) = 2^{-1/4} T_{\text{eff}}$.

Why $T_{\rm eq}$ is not enough: a myriad of effects

Without data, an exoplanet is labeled by its equilibrium temperature, computable without knowing anything about its atmosphere. The Solar System shows how misleading that can be: Earth's $T_{\rm eq}$ is below freezing, yet it has liquid water; Venus's $T_{\rm eq}$ is barely higher, yet its surface is about 700 K. Real atmospheres contain many opacity sources that warm or cool them (Heng, Ch. 4):

- **Greenhouse warming:** gases such as H_2O, CO_2 and CH_4 absorb the planet's *thermal* emission.
- **Anti-greenhouse cooling:** enhanced absorption, or scattering, of *starlight* high in the atmosphere.
- **Scattering greenhouse:** thermal emission is scattered back toward the planet.
- **Aerosols and clouds** combine all of these, and can cool or heat depending on their properties and where they sit.

The aim, following Pierrehumbert, is to understand this richness with simple, clear models that isolate one effect at a time — before, or instead of, a full simulation.

Convection versus Radiation: Which process carries the energy?

Energy can be transported by **radiation, convection, or conduction**. In the relatively dense lower atmospheres, thermal conduction is negligible. Energy takes the **path of least resistance**: if the opacity is low, then the energy is most likely to be transported by photons; if the opacity is high, radiation is impeded and energy is more likely to be transported by convection.

How the temperature gradient determines whether or not convection occurs:
Stability of a displaced parcel. Lift a parcel slightly: it expands *adiabatically* to the ambient pressure, keeping its own temperature and density. (adiabatic means no heat is exchanged with the surroundings.)

- if it becomes colder (denser) than its new surroundings: it sinks back. The atmosphere is **stable** against convection.
- if it becomes warmer (lighter) than its surroundings: buoyancy keeps it rising. The atmosphere is **unstable**; convection occurs.

A situation is **stable** if, after a disturbance, the system returns to its original state. Which case occurs is decided by comparing two temperature gradients (next slide).

The criterion for convection

Two temperature gradients compete:

- the **adiabatic** gradient $(\mathrm{d}T/\mathrm{d}z)_{\mathrm{ad}}$, followed by the rising parcel;
- the **atmospheric** gradient $(\mathrm{d}T/\mathrm{d}z)_{\mathrm{atm}}$ of the surroundings.

Start the parcel at height z with the ambient temperature T_0 and lift it by Δz. Both gradients are negative where T falls with height, so

$$T_{\mathrm{parcel}} = T_0 - \left|\frac{\mathrm{d}T}{\mathrm{d}z}\right|_{\mathrm{ad}} \Delta z, \qquad T_{\mathrm{atm}} = T_0 - \left|\frac{\mathrm{d}T}{\mathrm{d}z}\right|_{\mathrm{atm}} \Delta z.$$

The parcel has the ambient pressure, so by the ideal gas law ($\rho = \mu_m m_{\mathrm{H}} P/kT$, same μ_m) **the warmer gas is lighter**. The parcel keeps rising if $T_{\mathrm{parcel}} > T_{\mathrm{atm}}$, so **the criterion for convection is a criterion for buoyancy**:

$$\boxed{\left|\frac{\mathrm{d}T}{\mathrm{d}z}\right|_{\mathrm{atm}} > \left|\frac{\mathrm{d}T}{\mathrm{d}z}\right|_{\mathrm{ad}}}$$

(Schwarzschild criterion; Seager Eq. 9.23).

- Unstable: the atmosphere cools with height *faster* than the adiabat; the adiabat is the "shallower" gradient.
- Stable: the atmosphere cools more slowly than the adiabat, is isothermal, or warms with height (an inversion). Note that $T_{\mathrm{atm}} = T_0 + |\mathrm{d}T/\mathrm{d}z|_{\mathrm{atm}} \Delta z$ when temperature increases with height.

Stable and unstable profiles

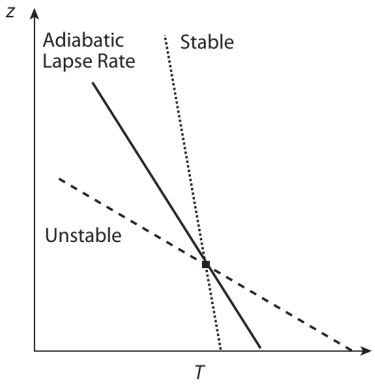


Figure 2: Profiles stable (dotted) or unstable (dashed) against convection, relative to the adiabat. Seager (2010), Fig. 9.6.

In the left figure (Fig. 9.6, T horizontal, z vertical), starting from a common point on the adiabat (solid line):

- a profile that loses *less* temperature per unit height than the adiabat: a lifted parcel ends up colder and denser than its surroundings and sinks back: **stable**;
- a profile that loses *more* temperature per unit height than the adiabat: the lifted parcel is warmer and lighter and keeps rising: **unstable**.

An unstable profile cannot persist: convection mixes the layer and drives its gradient back towards the adiabat. Convective regions therefore follow the adiabat closely.

To decide *when* a layer is unstable we need the actual sizes of the two gradients (next slide).

When is the adiabatic gradient small, or the radiative gradient large?

Which $(\mathrm{d}T/\mathrm{d}z)_{\mathrm{atm}}$ goes into the criterion? We ask whether a layer in which *radiation alone* carries the flux would be stable. Its gradient is then the radiative-equilibrium one, $(\mathrm{d}T/\mathrm{d}z)_{\mathrm{atm}} \approx (\mathrm{d}T/\mathrm{d}z)_{\mathrm{rad}}$. If that gradient is unstable, convection sets in and the actual gradient relaxes towards the adiabat.

The two gradients to compare (Seager Eqs. 9.24–9.25):

$$\left(\frac{\mathrm{d}T}{\mathrm{d}z}\right)_{\mathrm{ad}} = -\frac{g}{c_p}, \qquad \left(\frac{\mathrm{d}T}{\mathrm{d}z}\right)_{\mathrm{rad}} = -\frac{3}{16}\frac{\kappa F_{\mathrm{int}}}{\sigma_{\mathrm{SB}} T^3} \quad \Longrightarrow \quad \text{convection if} \quad \frac{3\kappa F_{\mathrm{int}}}{16\sigma_{\mathrm{SB}} T^3} > \frac{g}{c_p}.$$

The adiabat is derived later. The radiative gradient is the Rosseland diffusion result, with κ the mean extinction per unit length (α_R).

- **Small adiabatic gradient = large c_p.** c_p is the heat capacity, the energy needed to heat unit mass of gas by 1 K at constant pressure. Where hydrogen ionizes, the added heat goes into removing electrons rather than into the kinetic energy of the gas, so c_p becomes large and g/c_p small: ionization zones of stars, and the hot interiors of giant planets like Jupiter, where hydrogen starts to ionize above ~ 4000 K.
- **Large radiative gradient = large F_{int} or large κ.** The diffusion formula holds only deep in the atmosphere, but it shows the trend. Large fluxes drive the convective interiors of giant planets; in the atmospheres the flux alone is usually not large enough, and the **opacity** is the main driver (next slide).

Seager gives the units of c_p as $\mathrm{J\,kg^{-1}\,s^{-1}}$; they are $\mathrm{J\,kg^{-1}\,K^{-1}}$.

Convection in planets

- **Opacity drives convection.** Opacity grows with the density of absorbers, so the deep, optically thick layers are almost always convective, while radiation dominates higher up. Opacity can also change with T and P as the composition changes.
- **Jupiter:** a radiative zone in the upper atmosphere and a convective zone in the lower atmosphere — but perhaps another radiative zone underneath, where the mean opacity has a minimum ($T \approx 1300$–$1800\,\mathrm{K}$; Guillot et al. 1994).
- **Hot Jupiters:** strong irradiation makes the dayside temperature profile nearly isothermal or even inverted, so the radiative zone extends *deep* into the atmosphere.
- **Rocky planets with thin atmospheres:** most visible-wavelength starlight is absorbed at the planet surface, which becomes much hotter than the air above. The large upward flux from the ground makes the radiative gradient steep, and a **convective layer** forms just above the surface — on Earth most of the troposphere.
- **Inversions are very stable:** a lifted parcel cools as it expands while its surroundings get warmer with height, so it sinks back strongly. Stratospheres do not convect. Most of the solar system planets have stratospheres, heated by UV absorption in ozone (Earth) or hydrocarbons (Jupiter, Saturn, Titan).

Radiative equilibrium

If energy is carried only by radiation, no energy is created or destroyed in a layer: what enters a volume must leave it. Equivalently, the net flux is constant,

$$\frac{\mathrm{d}F(\tau)}{\mathrm{d}\tau} = 0.$$

Total absorption per volume is $\int\!\!\int \kappa_\nu I_\nu \,\mathrm{d}\Omega\,\mathrm{d}\nu = 4\pi \int \kappa_\nu J_\nu \,\mathrm{d}\nu$ $(\kappa_\nu J_\nu = (1/4\pi)\int \kappa_\nu I_\nu \mathrm{d}\Omega)$; with isotropic emission and $\varepsilon_\nu = \kappa_\nu S_\nu$, total emission is $4\pi \int \kappa_\nu S_\nu \,\mathrm{d}\nu$. Hence

$$\boxed{\int_0^\infty \kappa_\nu \,(J_\nu - S_\nu)\,\mathrm{d}\nu = 0}$$

- Absorption and emission balance *in total*, not at each frequency: a layer can absorb in the visible and emit in the infrared.
- The constraint involves J_ν, so it must be solved *together* with the transfer equation. The opacity's frequency dependence enters directly, and an inversion can arise naturally from the solution.
- The thermal (infrared) flux at the top is $\sigma T_{\mathrm{int}}^4 + \mu_0 I_0$: the interior heat plus the reradiated starlight. Interior heat comes from slow loss of formation energy (Jupiter emits about 1.7 times what it absorbs) or radioactivity (Earth); for Earth and close-in planets the stellar term dominates.

No general analytic solution exists; the **gray** atmosphere gives one.

Gray atmosphere heated from below

Assume LTE, no scattering, and frequency-independent opacity (gray). The energy comes from below only: from the interior of a giant planet ($T_{\rm int} = T_{\rm eff}$), or from a ground heated by sunlight that the atmosphere does not absorb ($T_{\rm int} = T_s$).

Step 1: radiative equilibrium in the gray case. With the frequency-integrated quantities $J(\tau) = \int_0^\infty J_\nu \,\mathrm{d}\nu$, $I(\tau,\mu) = \int_0^\infty I_\nu \,\mathrm{d}\nu$, $S(\tau) = \int_0^\infty S_\nu \,\mathrm{d}\nu$, and $\kappa_\nu = \kappa$, the condition $\int \kappa_\nu (J_\nu - S_\nu)\,\mathrm{d}\nu = 0$ becomes $J = S$. In LTE, $S = B$ integrated over frequency, so

$$J(\tau) = S(\tau) = B\big(T(\tau)\big) = \frac{\sigma T^4}{\pi}.$$

Step 2: the gray transfer equation. Integrating $\mu\,\mathrm{d}I_\nu/\mathrm{d}\tau = I_\nu - S_\nu$ over frequency and using $S = J$,

$$\mu \frac{\mathrm{d}I(\tau,\mu)}{\mathrm{d}\tau} = I(\tau,\mu) - J(\tau).$$

The goal is to show $T^4 \approx \frac{3}{4} T_{\rm int}^4 (\tau + \frac{2}{3})$. The method (Seager follows Hansen 2008) is to take **moments**: multiply by powers of μ and integrate over solid angle, $\mathrm{d}\Omega = \mathrm{d}\mu\,\mathrm{d}\phi$ with $-1 \le \mu \le 1$, $0 \le \phi < 2\pi$.

Gray atmosphere heated from below: moments

Recall the moments $J = \frac{1}{4\pi}\oint I\,\mathrm{d}\Omega$, $F = \oint I\mu\,\mathrm{d}\Omega$, $K = \frac{1}{4\pi}\oint I\mu^2\,\mathrm{d}\Omega$.

Step 3: zeroth moment ($\times \frac{1}{4\pi}\oint \mathrm{d}\Omega$):

$$\frac{1}{4\pi}\frac{\mathrm{d}F}{\mathrm{d}\tau} = J - J = 0 \quad \Longrightarrow \quad F(\tau) = F = \text{const.}$$

This is the flux constancy of radiative equilibrium. All the flux comes from below, so $F = \sigma T_{\text{int}}^4$ at every level.

Step 4: first moment ($\times \frac{1}{4\pi}\oint \mu\,\mathrm{d}\Omega$; the term $J\oint \mu\,\mathrm{d}\Omega$ vanishes because J is isotropic):

$$\frac{\mathrm{d}K}{\mathrm{d}\tau} = \frac{F}{4\pi} \quad \Longrightarrow \quad K(\tau) = \frac{F}{4\pi}\,\tau + c,$$

with an integration constant c, since F does not depend on τ.

Step 5: close the system with the Eddington approximation $K = \frac{1}{3}J$ (Lecture 7):

$$J(\tau) = \frac{3F}{4\pi}\,\tau + 3c.$$

The moment equations alone cannot fix c; it needs a boundary condition at the top of the atmosphere.

Gray atmosphere heated from below: the boundary condition

Step 6: the emergent flux. At $\tau = 0$ nothing comes in from above, and the outgoing intensity is the source function attenuated along the slant path, $I(0,\mu) = \int_0^\infty S(\tau)\,\mathrm{e}^{-\tau/\mu}\,\mathrm{d}\tau/\mu$. Then $F(0) = 2\pi\int_0^1 I(0,\mu)\,\mu\,\mathrm{d}\mu$ gives

$$F(0) = 2\pi \int_0^1 \int_0^\infty S(\tau)\,\mathrm{e}^{-\tau/\mu}\,\mathrm{d}\tau\,\mathrm{d}\mu.$$

Insert $S = J = a\tau + b$ with $a = 3F/4\pi$, $b = 3c$, and use $\int_0^\infty \tau\,\mathrm{e}^{-\tau/\mu}\mathrm{d}\tau = \mu^2$, $\int_0^\infty \mathrm{e}^{-\tau/\mu}\mathrm{d}\tau = \mu$:

$$F(0) = 2\pi\int_0^1 \left(a\mu^2 + b\mu\right)\mathrm{d}\mu = 2\pi\left(\frac{a}{3} + \frac{b}{2}\right) = \frac{F}{2} + \pi b.$$

Flux constancy requires $F(0) = F = \sigma T_{\mathrm{int}}^4$, so $\pi b = F/2$, i.e. $3c = F/2\pi = \sigma T_{\mathrm{int}}^4/2\pi$.

Step 7: the temperature profile. Now $J = \frac{3F}{4\pi}\left(\tau + \frac{2}{3}\right)$. Setting $J = \sigma T^4/\pi$ and $F = \sigma T_{\mathrm{int}}^4$:

$$\frac{\sigma T^4}{\pi} = \frac{3\sigma T_{\mathrm{int}}^4}{4\pi}\left(\tau + \frac{2}{3}\right) \quad\Longrightarrow\quad \boxed{T^4 \approx \frac{3}{4}\,T_{\mathrm{int}}^4\left(\tau + \frac{2}{3}\right)}$$

The "$\approx$" comes from the Eddington closure $K = J/3$; the exact gray solution replaces $2/3$ by a slowly varying function $q(\tau)$, the Hopf function, with $q(0) = 1/\sqrt{3} \approx 0.577$ and $q(\infty) \approx 0.710$.

Gray atmosphere heated from below: what it tells us

$$T^4 \approx \frac{3}{4} T_{\rm int}^4 \left(\tau + \frac{2}{3} \right)$$

- **Top of the atmosphere:** $T(0)^4 = T_{\rm int}^4/2$, the same $2^{-1/4}$ ratio as the simple greenhouse (Lecture 7).
- **Gradient:** differentiating with $\mathrm{d}\tau = -\kappa\,\mathrm{d}z$,

$$\frac{\mathrm{d}T}{\mathrm{d}z} = -\frac{3}{16}\,\kappa\,\frac{T_{\rm int}^4}{T^3},$$

  the radiative-diffusion gradient. Large flux or large opacity gives a steep gradient.
- With the opacity's T, P dependence and hydrostatic equilibrium, $T(\tau)$ becomes a T–P profile.

Gray atmosphere heated from above and below: why two bands?

The profile above is for an atmosphere **heated from below** only. It can represent

- a giant planet whose **interior energy** heats the atmosphere, or
- a rocky planet with a surface, where the incident starlight passes through a **transparent** atmosphere, is absorbed by the ground, and is reradiated upward.

In general, however, **starlight is absorbed throughout the atmosphere**. We keep the same simplifications (gray, LTE, radiative equilibrium, no scattering) and add heating from above (Seager §9.6.3, following Hansen 2008).

The key point: the star is always hotter than the planet. Stellar radiation is therefore absorbed at **visible** wavelengths and reradiated at **infrared** wavelengths, where molecular absorption usually dominates. This suggests two intensities:

- I_{vis}: radiation is only **absorbed**, not emitted;
- I_{ir}: all the absorbed energy is **reradiated**.

The total intensity is $I_{\text{vis}} + I_{\text{ir}}$, and the same addition rule holds for J, F, and K.

Why separate them: radiation is absorbed and emitted at *different altitudes*. Earth's cloudless atmosphere is transparent in the visible (we see the stars at night), so most visible light is absorbed by the ground; at most infrared wavelengths, apart from a few narrow windows, the night sky is opaque.

Heated from above and below: Setup and visible beam

Step 1: radiative equilibrium in two gray bands (Seager §9.6.3; Hansen 2008). With constant opacities $\kappa_{\rm vis}$ and $\kappa_{\rm ir}$, and no visible emission ($S_{\rm vis} = 0$), the equilibrium condition $\int \kappa_\nu (J_\nu - S_\nu)\,\mathrm{d}\nu = 0$ balances visible plus infrared absorption against infrared emission:

$$\kappa_{\rm vis} J_{\rm vis} + \kappa_{\rm ir} J_{\rm ir} = \varepsilon_{\rm ir} = \kappa_{\rm ir} S_{\rm ir} \quad \Longrightarrow \quad \boxed{S_{\rm ir} = J_{\rm ir} + \gamma J_{\rm vis}, \qquad \gamma \equiv \frac{\kappa_{\rm vis}}{\kappa_{\rm ir}}}$$

Write $\tau \equiv \tau_{\rm ir}$ so that $\mathrm{d}\tau_{\rm vis} = \gamma\,\mathrm{d}\tau$.

Step 2: the downward stellar beam. With $S_{\rm vis} = 0$, the visible transfer equation is simply $\mu\,\mathrm{d}I_{\rm vis}/\mathrm{d}\tau_{\rm vis} = I_{\rm vis}$. Collimated starlight enters from direction (θ_0, ϕ_0) with $\mu_0 = \cos\theta_0$ (where the positive z direction is upward):

$$I_{\rm vis}(\tau, \theta, \phi) = I_0\, \mathrm{e}^{-\gamma\tau/\mu_0}\, \delta(\cos\theta + \mu_0)\, \delta(\phi - \phi_0) \quad \big[\text{or } I(\theta,\phi) = I_0\, \delta(\theta - \theta_0)\, \delta(\phi - \phi_0)\big].$$

Its mean intensity and downward flux follow immediately by integrating over $\mathrm{d}\Omega$:

$$J_{\rm vis}(\tau) = \frac{I_0}{4\pi}\, \mathrm{e}^{-\gamma\tau/\mu_0}, \qquad F_{\rm vis}(\tau) = -\mu_0 I_0\, \mathrm{e}^{-\gamma\tau/\mu_0} \quad \text{(downward)}.$$

Heated from above and below: Infrared moments

Step 3: the infrared transfer equation. Substituting $S_{\mathrm{ir}} = J_{\mathrm{ir}} + \gamma J_{\mathrm{vis}}$ into the gray IR transfer equation gives

$$\mu \frac{\mathrm{d}I_{\mathrm{ir}}}{\mathrm{d}\tau} = I_{\mathrm{ir}} - S_{\mathrm{ir}} = I_{\mathrm{ir}} - J_{\mathrm{ir}} - \gamma J_{\mathrm{vis}}.$$

Step 4: zeroth moment ($\times \frac{1}{4\pi} \oint \mathrm{d}\Omega$). Since J_{ir} and J_{vis} are isotropic,

$$\frac{1}{4\pi} \frac{\mathrm{d}F_{\mathrm{ir}}}{\mathrm{d}\tau} = J_{\mathrm{ir}} - S_{\mathrm{ir}} = -\gamma J_{\mathrm{vis}} = -\frac{\gamma I_0}{4\pi} \mathrm{e}^{-\gamma\tau/\mu_0}.$$

Integrating over τ (or directly from total flux conservation $F_{\mathrm{ir}} + F_{\mathrm{vis}} = F_{\mathrm{int}} = \sigma T_{\mathrm{int}}^4 = \mathrm{const}$),

$$F_{\mathrm{ir}}(\tau) = -F_{\mathrm{vis}}(\tau) + F_{\mathrm{int}} = \mu_0 I_0 \mathrm{e}^{-\gamma\tau/\mu_0} + \sigma T_{\mathrm{int}}^4.$$

Step 5: first moment ($\times \frac{1}{4\pi} \oint \mu \, \mathrm{d}\Omega$). The isotropic terms $(J_{\mathrm{ir}} + \gamma J_{\mathrm{vis}}) \oint \mu \, \mathrm{d}\Omega$ vanish:

$$\frac{\mathrm{d}K_{\mathrm{ir}}}{\mathrm{d}\tau} = \frac{1}{4\pi} F_{\mathrm{ir}}(\tau) = \frac{1}{4\pi} \left(\mu_0 I_0 \mathrm{e}^{-\gamma\tau/\mu_0} + \sigma T_{\mathrm{int}}^4 \right).$$

Integrating over τ and applying the Eddington closure $K_{\mathrm{ir}} = \frac{1}{3} J_{\mathrm{ir}}$ gives

$$J_{\mathrm{ir}}(\tau) = 3K_{\mathrm{ir}}(\tau) = \frac{3}{4\pi} \left(\sigma T_{\mathrm{int}}^4 \, \tau - \frac{\mu_0^2}{\gamma} I_0 \mathrm{e}^{-\gamma\tau/\mu_0} \right) + 3c.$$

Heated from above and below: Boundary condition

Step 6: fix the integration constant c from the emergent IR flux. At the top of the atmosphere ($\tau = 0$), no IR radiation enters from space, and the outgoing intensity is $I_{\rm ir}(0,\mu) = \int_0^\infty S_{\rm ir}(\tau)\,\mathrm{e}^{-\tau/\mu}\,\mathrm{d}\tau/\mu$. The upward IR flux at $\tau = 0$ is therefore (Seager Eq. 9.58)

$$F_{\rm ir}(0) = 2\pi \int_0^1 I_{\rm ir}(0,\mu)\,\mu\,\mathrm{d}\mu = 2\pi \int_0^1 \int_0^\infty S_{\rm ir}(\tau)\,\mathrm{e}^{-\tau/\mu}\,\mathrm{d}\tau\,\mathrm{d}\mu.$$

Substitute $S_{\rm ir}(\tau) = J_{\rm ir}(\tau) + \gamma J_{\rm vis}(\tau)$ using Steps 2 and 5:

$$S_{\rm ir}(\tau) = \frac{3\sigma T_{\rm int}^4}{4\pi}\,\tau + 3c + \frac{I_0}{4\pi}\left(\gamma - \frac{3\mu_0^2}{\gamma}\right)\mathrm{e}^{-\gamma\tau/\mu_0}.$$

Carry out the τ-integrals with $\int_0^\infty \tau\,\mathrm{e}^{-\tau/\mu}\mathrm{d}\tau = \mu^2$, $\int_0^\infty \mathrm{e}^{-\tau/\mu}\mathrm{d}\tau = \mu$, and $\int_0^\infty \mathrm{e}^{-\tau(1/\mu+\gamma/\mu_0)}\mathrm{d}\tau = \mu\left(1+\frac{\gamma\mu}{\mu_0}\right)^{-1}$:

$$F_{\rm ir}(0) = \frac{1}{2}\sigma T_{\rm int}^4 + 3\pi c + \frac{I_0}{2}\left(\gamma - \frac{3\mu_0^2}{\gamma}\right)\int_0^1 \frac{\mu^2\,\mathrm{d}\mu}{1+\gamma\mu/\mu_0}.$$

Equating this to the required top flux from Step 4, $F_{\rm ir}(0) = \mu_0 I_0 + \sigma T_{\rm int}^4$, determines $3c$.

Heated from above and below: The temperature profile $T(\tau)$

Step 7: solve for $T(\tau)$ via LTE ($S_{\rm ir} = \sigma T^4/\pi$). With $I_0 \equiv \sigma T_0^4$, the constant $3c$ can be fixed in two slightly different ways:

- **Eddington boundary condition** $J_{\rm ir}(0) = F_{\rm ir}(0)/2\pi$ (at $\tau = 0$, $I_{\rm ir} = 0$ for $\mu < 0$ and $I_{\rm ir} \approx I^+$ is isotropic for $\mu > 0$, so $J_{\rm ir}(0) = \frac{1}{2}\int_0^1 I^+ \mathrm{d}\mu = \frac{1}{2} I^+$ and $F_{\rm ir}(0) = 2\pi \int_0^1 I^+ \mu \, \mathrm{d}\mu = \pi I^+$): this gives $3c = \frac{\sigma}{2\pi}\left[T_{\rm int}^4 + \mu_0 T_0^4 \left(1 + \frac{3\mu_0}{2\gamma}\right)\right]$, and $\frac{\sigma T^4}{\pi} = S_{\rm ir} = J_{\rm ir} + \gamma J_{\rm vis}$ yields

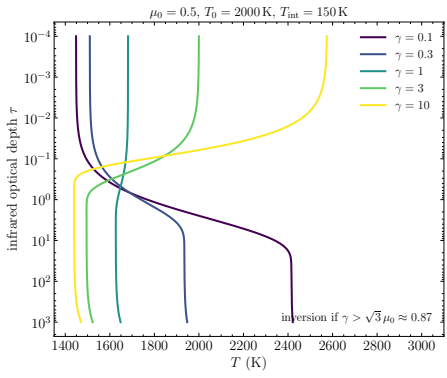
$$T^4(\tau) = \frac{3}{4} T_{\rm int}^4 \left(\tau + \frac{2}{3}\right) + \frac{\mu_0 T_0^4}{4}\left[2 + \frac{3\mu_0}{\gamma} + \left(\frac{\gamma}{\mu_0} - \frac{3\mu_0}{\gamma}\right) \mathrm{e}^{-\gamma\tau/\mu_0}\right]$$

- **Exact emergent-flux integral** (Seager Eqs. 9.58–9.62; Hansen 2008): evaluating $\int_0^1 \frac{\mu^2 \, \mathrm{d}\mu}{1 + \gamma\mu/\mu_0}$ explicitly in Step 6 gives

$$\begin{aligned} T^4(\tau) \approx \frac{3}{4} T_{\rm int}^4 \left(\tau + \frac{2}{3}\right) + \frac{3}{4}\mu_0 T_0^4 \left[\frac{2}{3} + \frac{\mu_0}{\gamma} - \frac{\mu_0}{\gamma} \mathrm{e}^{-\gamma\tau/\mu_0}\right] \\ - \frac{1}{2} T_0^4 \left(\frac{\gamma}{\mu_0} - \frac{3\mu_0}{\gamma}\right) \mu_0^2 \left[\frac{1}{2} - \frac{\mu_0}{\gamma} + \frac{\mu_0^2}{\gamma^2} \ln\left(1 + \frac{\gamma}{\mu_0}\right)\right] \end{aligned}$$

Note: Seager's Eq. (9.60) sets $\sigma T^4/\pi = J_{\rm ir}(\tau)$ in the last step, omitting the $+\gamma J_{\rm vis}(\tau)$ term from $S_{\rm ir} = J_{\rm ir} + \gamma J_{\rm vis}$ (Eq. 9.48); keeping it adds the $\frac{\gamma}{\mu_0} \mathrm{e}^{-\gamma\tau/\mu_0}$ term.

Irradiated gray τ-T profiles



The solution with the Eddington boundary condition ($J_{\mathrm{ir}}(0) = F_{\mathrm{ir}}(0)/2\pi$) for several γ. Script `scripts/make_irradiated_gray_profiles.py`.

- **Deep layers** ($\gamma\tau/\mu_0 \gg 1$): the starlight has been absorbed; $T^4 \to \frac{3}{4} T_{\mathrm{int}}^4 \tau + \mathrm{const}$ — the interior-heated profile, nearly isothermal when T_{int} is small.
- **Visible transparent** ($\gamma \to 0$): the profile tends to $\frac{3}{4}(T_{\mathrm{int}}^4 + \mu_0 T_0^4)(\tau + \frac{2}{3})$, as if all the flux came from below — the ground-heated case.
- **Thermal inversion** (T increases upward, as $\tau \to 0$) occurs when the coefficient of $\mathrm{e}^{-\gamma\tau/\mu_0}$ is positive:

$$\frac{\gamma}{\mu_0} - \frac{3\mu_0}{\gamma} > 0 \quad \Longleftrightarrow \quad \gamma > \sqrt{3}\,\mu_0 \quad (\kappa_{\mathrm{vis}} > \sqrt{3}\,\mu_0\,\kappa_{\mathrm{ir}}).$$

  Physically: strong optical absorbers (TiO/VO, metals, ozone, hazes) trap starlight high in the atmosphere (large κ_{vis}), where the gas cannot cool efficiently in the infrared (small κ_{ir}), forcing the upper layers to heat up.
- Smaller μ_0 (grazing starlight, near the terminator) deposits the energy higher and lowers the deep temperature.

Irradiated gray T–P profiles

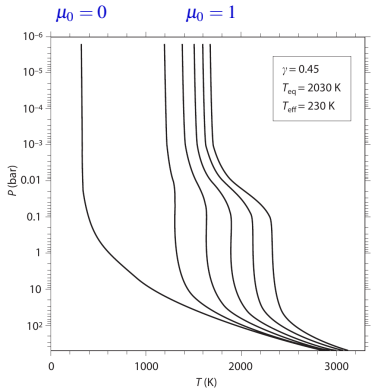


Figure 3: Gray-atmosphere T–P profiles for $\mu_0 = 0.0$–1.0 (left to right), $\gamma = 0.45$. Seager (2010), Fig. 9.7, after Hansen (2008).

- Hydrostatic equilibrium ($d\tau = \kappa_m \, dP/g$) converts $T(\tau)$ into a T–P profile.
- **Three altitude regions** (here $\gamma = 0.45 < \sqrt{3}\,\mu_0$, so no inversion):
  1. **Top ($P \lesssim 10^{-3}$ bar, $\tau \ll 1$):** optically thin $\rightarrow$ **isothermal** skin layer.
  2. **Middle (10^{-3}–10 bar):** starlight is absorbed around $\tau_{\rm vis} = \gamma\tau/\mu_0 \sim 1$ (T rises), then settles into a second **isothermal** zone once $\tau_{\rm vis} \gg 1$.
  3. **Deep ($P \gtrsim 10$ bar, $\tau \gg 1$):** interior flux ($T_{\rm int}$) dominates $\rightarrow$ T rises with depth ($T^4 \propto \tau$).
- **Incident angle** $\mu_0 = \cos\theta_0$ (left $\mu_0 = 0$ to right $\mu_0 = 1$): more vertical starlight brings larger flux ($\propto \mu_0$) and penetrates deeper, warming all levels.
- Widely used as the **initial guess** for full radiative-convective models.

The Adiabatic Temperature Profile: Convection basics

Convection transports energy by **bulk vertical motion**: heat flows from hot to cold regions with the moving gas. It is driven by the temperature gradient and enabled by gravity (buoyancy).

- A parcel heated from below becomes lighter than its surroundings and rises. At lower ambient pressure it expands and releases heat; now cooler, it sinks, is reheated, and the cycle repeats.
- Everyday examples: boiling water, lava lamps, the shimmering air above hot pavement.
- The Sun's **granulation**: bright centers are rising columns of hot gas, dark lanes are cooled gas sinking; cells are ~ 1000 km across.

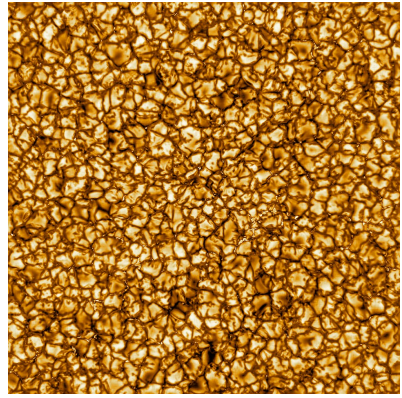


Figure 4: Solar granulation imaged by the Daniel K. Inouye Solar Telescope (NSO/AURA/NSF). Each cell is ~ 1000 km across.

Thermodynamics of an Ideal Gas: Internal Energy

- **First Law of Thermodynamics** (energy conservation):

Heat added to a system = Change in internal energy + Work done on surroundings

$$\boxed{\mathrm{d}Q = \mathrm{d}U + P\,\mathrm{d}V}$$

- **Internal energy per particle** of an ideal gas (N = number of particles):

$U/N = \frac{3}{2}k_{\mathrm{B}}T$	for a monatomic gas	(translation about 3 axes)
$U/N = \frac{5}{2}k_{\mathrm{B}}T$	for a diatomic gas	(+ rotation about 2 axes)
$U/N = 3k_{\mathrm{B}}T$	for a polyatomic gas	(+ rotation about 3 axes)

An ideal gas is a theoretical gas composed of many randomly moving point particles whose only interactions are perfectly elastic collisions (no viscosity or heat conduction).

- **In general**, the internal energy per particle is

$$\boxed{U/N = \frac{f}{2}k_{\mathrm{B}}T \qquad (f = \text{degrees of freedom})}$$

At high temperatures, molecules gain access to an increasing number of vibrational degrees of freedom as they start to bend and stretch.

Thermodynamics of an Ideal Gas: Specific Heat Capacities

- **Ideal gas law** (equation of state for a Maxwellian distribution): $\boxed{PV = Nk_\mathrm{B}T}$ or $P = \dfrac{N}{V}k_\mathrm{B}T$.
- **Specific heat capacity** is the heat energy required to raise the temperature of a material per unit mass (M = total mass, $m = M/N = \mu_m m_\mathrm{H}$ = mass per particle, $R_s = k_\mathrm{B}/m$ = specific gas constant):
  - **at constant volume** ($\mathrm{d}V = 0$):

$$c_v \equiv \frac{1}{M}\left(\frac{\partial Q}{\partial T}\right)_V = \frac{1}{M}\left(\frac{\partial U}{\partial T}\right)_V \quad \Longrightarrow \quad \boxed{c_v = \frac{f}{2}\frac{k_\mathrm{B}}{m} = \frac{f}{2}R_s}$$

  - **at constant pressure** ($\mathrm{d}P = 0$):

$$c_p \equiv \frac{1}{M}\left(\frac{\partial Q}{\partial T}\right)_P = \frac{1}{M}\left(\frac{\partial U}{\partial T}\right)_P + \frac{P}{M}\left(\frac{\partial V}{\partial T}\right)_P = \frac{1}{M}\frac{f}{2}Nk_\mathrm{B} + \frac{P}{M}\frac{Nk_\mathrm{B}}{P}$$

$$\Longrightarrow \quad \boxed{c_p = \frac{f+2}{2}\frac{k_\mathrm{B}}{m} = c_v + \frac{k_\mathrm{B}}{m} = c_v + R_s}$$

- **Ratio of specific heat capacities** (**adiabatic index** γ_ad; $c_p > c_v$ since part of the added heat at constant P goes into $P\,\mathrm{d}V$ expansion work):

$$\boxed{\gamma_\mathrm{ad} \equiv \frac{c_p}{c_v} = \frac{f+2}{f}} \quad \Longrightarrow \quad \gamma_\mathrm{ad} = \begin{cases} 5/3 & \text{monatomic } (f=3) \\ 7/5 & \text{diatomic } (f=5) \\ 4/3 & \text{polyatomic } (f=6) \end{cases}$$

Take-home messages

Toy greenhouse models (§9.5). Energy balance gives $T_s = [2/(2-\alpha)]^{1/4} T_e$ for a single leaky infrared-absorbing layer ($\alpha \simeq 0.78$ reproduces Earth's $T_s \simeq 288$ K) and $T_s = (n+1)^{1/4} T_e$ for n opaque layers ($n \simeq 100$ for Venus).

Radiative vs. convective transport (§9.4). Radiative equilibrium requires $\mathrm{d}F_{\mathrm{rad}}/\mathrm{d}\tau = 0$, or $\int \kappa_\nu (J_\nu - S_\nu)\,\mathrm{d}\nu = 0$. When high opacity or large flux makes the radiative temperature gradient steeper than the adiabatic gradient ($|\mathrm{d}T/\mathrm{d}z|_{\mathrm{rad}} > \Gamma_{\mathrm{ad}}$, Schwarzschild criterion), convection takes over.

Gray & irradiated two-band T–P profiles (§9.6). An unirradiated gray Eddington atmosphere obeys $T^4(\tau) = \frac{3}{4} T_{\mathrm{int}}^4 (\tau + \frac{2}{3})$. Adding incident starlight with visible-to-infrared opacity ratio $\gamma = \kappa_{\mathrm{vis}}/\kappa_{\mathrm{ir}}$ produces a **greenhouse effect** when $\gamma < \sqrt{3}\,\mu_0$ and a **thermal inversion** when $\gamma > \sqrt{3}\,\mu_0$.

Ideal gas thermodynamics (§9.7.1). For an ideal gas with f degrees of freedom, $U/N = (f/2) k_{\mathrm{B}} T$, $c_v = (f/2) R_s$, $c_p = c_v + R_s = [(f+2)/2] R_s$, and $\gamma_{\mathrm{ad}} = c_p/c_v = (f+2)/f$.

Next (Lecture 9). Dry and moist adiabatic temperature gradients, 1D radiative–convective T–P models, collision-induced absorption (CIA), and parametric temperature retrieval.

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